QUESTIONS FOR DISCUSSION

1. Verify that the Brinkman, Biot, Prandtl, and Grashof numbers are dimensionless.
2. To what problem in electrical circuits is the addition of thermal resistances analogous?
3. What is the coefficient of volume expansion for an ideal gas? What is the corresponding expression for the Grashof number?
4. What might be some consequences of large temperature gradients produced by viscous heating in viscometry, lubrication, and plastics extrusion?
5. In §10.8 would there be any advantage to choosing the dimensionless temperature and dimensionless axial coordinate to be $\Theta = (T - T_1)/T_1$ and $\zeta = z/R$?
6. What would happen in §9.9 if the fluid were water and $\overline{T}$ were 4°C?
7. Is there any advantage to solving Eq. 9.7-9 in terms of hyperbolic functions rather than exponential functions?
8. In going from Eq. 10.8-11 to Eq. 10.8-12 the axial conduction term was neglected with respect to the axial convection term. To justify this, put in some reasonable numerical values to estimate the relative sizes of the terms.
9. How serious is it to neglect the dependence of viscosity on temperature in solving forced convection problems? Viscous dissipation heating problems?
10. At steady state the temperature profiles in a laminated system appear thus: Which material has the higher thermal conductivity?

3.) Coeffifient of volume expansion: β

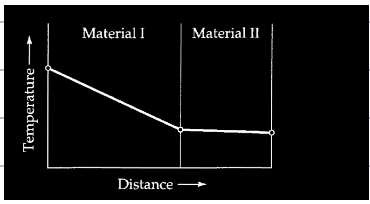
$$Gr_L = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2}$$

8.)

$$\rho\hat{C}_p v_z \frac{\partial T}{\partial z} = k\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{\partial^2 T}{\partial z^2}\right] + \mu\left(\frac{\partial v_z}{\partial r}\right)^2 + v_z\left[-\frac{\partial p}{\partial z} + \mu\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \rho g\right] \quad (10.8\text{-}11)$$

$$\rho\hat{C}_p v_{z,\max}\left[1 - \left(\frac{r}{R}\right)^2\right]\frac{\partial T}{\partial z} = k\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)\right] \quad (10.8\text{-}12)$$

10.)



Material 1 has the higher thermal Conductivity because it loses heat heat at a faster rate than it's Supplied insinuating the thermal Conductivity is high Since it can't retain temperature.

10A.5. Free convection velocity.

(a) Verify the expression for the average velocity in the upward-moving stream in Eq. 10.9-16.

(b) Evaluate $\overline{\beta}$ for the conditions given below.

(c) What is the average velocity in the upward-moving stream in the system described in Fig. 10.9-1 for air flowing under these conditions?

Pressure	1 atm
Temperature of the heated wall	100°C
Temperature of the cooled wall	20°C
Spacing between the walls	0.6 cm

Answer: 2.3 cm/s

a.) $\langle v_z \rangle = \frac{(\bar{\rho} g \bar{\beta} \Delta T) B^2}{48\mu}$ Stated in class didn't have to do

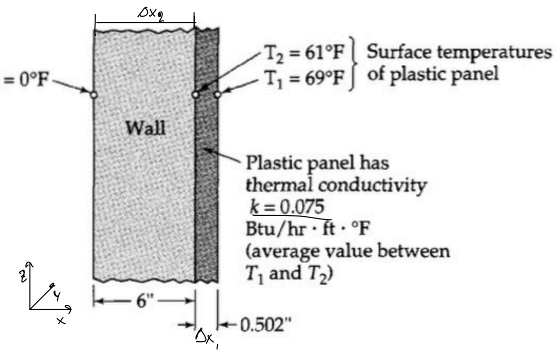


Fig. 10A.6. Determination of the thermal resistance of a wall.

10A.6. Insulating power of a wall (Fig. 10A.6). The "insulating power" of a wall can be measured by means of the arrangement shown in the figure. One places a plastic panel against the wall. In the panel two thermocouples are mounted flush with the panel surfaces. The thermal conductivity and thickness of the plastic panel are known. From the measured steady-state temperatures shown in the figure, calculate:

(a) The steady-state heat flux through the wall (and panel).

(b) The "thermal resistance" (wall thickness divided by thermal conductivity).

Answers: **(a)** 14.3 Btu/hr · ft²; **(b)** 4.2 ft² · hr · F/Btu

a.) Things to Learn from this: q = [heat flux] @ S.S. is Constant through Surfaces

$$q = -k\frac{dT}{dx} = -k\frac{T_2-T_1}{\Delta x_1} = -0.075\ \frac{Btu}{hr.ft.°F}\left(\frac{61°F-69°F}{0.502\,in \,\big|\, \frac{ft}{12\,in}}\right) = \boxed{14.34\ \frac{Btu}{hr\,ft^2}}$$

b.) $$R = \frac{\Delta x_2}{K_{12}} = \frac{0.5ft}{0.1176\ \frac{Btu}{hrft°F}} = \boxed{4.3\ \frac{hrft^2°F}{Btu}}$$

$$q = 14.34 = -K_{12}\frac{(0-61)}{0.5}$$

$$-K_{12} = \frac{14.34\cdot 0.5}{-61}\left(\frac{Btu}{hrft°F}\right)$$

$$K_{12} = 0.1176\ \frac{Btu}{hrft°F}$$

10B.1. Heat conduction from a sphere to a stagnant fluid. A heated sphere of radius R is suspended in a large, motionless body of fluid. It is desired to study the heat conduction in the fluid surrounding the sphere in the absence of convection.

(a) Set up the differential equation describing the temperature T in the surrounding fluid as a function of r, the distance from the center of the sphere. The thermal conductivity k of the fluid is considered constant.

(b) Integrate the differential equation and use these boundary conditions to determine the integration constants: at $r = R$, $T = T_R$; and at $r = \infty$, $T = T_\infty$.

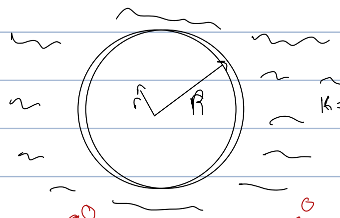
(c) From the temperature profile, obtain an expression for the heat flux at the surface. Equate this result to the heat flux given by "Newton's law of cooling" and show that a dimensionless heat transfer coefficient (known as the *Nusselt number*) is given by

$$\text{Nu} = \frac{hD}{k} = 2 \qquad \text{(10B.1-1)}$$

in which D is the sphere diameter. This well-known result provides the limiting value of Nu for heat transfer from spheres at low Reynolds and Grashof numbers (see §14.4).

(d) In what respect are the Biot number and the Nusselt number different?

a.)



K = constant

Assumptions:

$T = T(r)$ Steady state

$q = q(r) = q_r$

$q_\phi = q_\theta = 0$

$$\int_V dV \frac{\partial}{\partial t}\rho\left(\tfrac{1}{2}v^2+\Phi+u\right) = -\int_S dS\, \rho\left(\tfrac{1}{2}v^2+\Phi+u\right)(v\cdot n) - \int_S dS(q\cdot n) - \int_S dS(v\cdot\tau\cdot n) + \int_V dV \dot{S}$$

not a function of time @ S.S no work no energy generated

$$0 = -\int_S dS(q\cdot n)$$

$$0 = 4\pi r^2\left([q_r\, \cancel{q_\theta}\, \cancel{q_\phi}]_r - [q_r\, \cancel{q_\theta}\, \cancel{q_\phi}]_{r+\Delta r}\right)$$

$$0 = \lim_{\Delta r\to 0} \frac{-([4\pi r^2 q_r]_r - [4\pi r^2 q_r]_{r+\Delta r})}{4\pi r^2 \Delta r} = -\frac{1}{r^2}\lim_{\Delta r\to 0}\frac{[r^2 q_r]_r - [r^2 q_r]_{r+\Delta r}}{\Delta r}$$

$$0 = -\frac{1}{r^2}\frac{d(r^2 q_r)}{dr} \Rightarrow 0 = -\frac{1}{r^2}\frac{d\left(r^2\left(-K\frac{dT}{dr}\right)\right)}{dr} \Rightarrow 0 = \frac{K}{r^2}\frac{d}{dr}\left(r^2\frac{dT}{dr}\right)$$

b.) BC:

$T(R) = T_R$

$T(\infty) = T_\infty$

$$0 = \frac{d}{dr}(r^2 q_r) = C_1 = r^2 q_r \Rightarrow \frac{C_1}{r^2} = q_r, \quad q_r = -K\frac{dT}{dr}$$

$$\frac{C_1}{r^2} = -K\frac{dT}{dr} \Rightarrow \frac{dT}{dr} = -\frac{C_1}{Kr^2} \Rightarrow T(r) = \frac{C_1}{Kr} + C_2$$

$$T_R = \frac{C_1}{KR} + C_2 \qquad T(r) = \int T$$

$$T_\infty = \frac{C_1}{K(\infty)} + C_2 = C_2$$

$$T_R = \frac{C_1}{KR} + T_\infty$$

$$C_1 = (T_R - T_\infty)KR$$

$$T(r) = \frac{(T_R - T_\infty)KR}{Kr} + T_\infty = T(r) = \frac{(T_R - T_\infty)R}{r} + T_\infty$$

c.) $$q_r = -K\frac{dT}{dr}\Big|^{r=R} = -K\frac{d}{dr}\left(\frac{(T_R-T_\infty)R}{r} + T_\infty\right) = \frac{K(T_r - T_\infty)R}{r^2}\Big|^{r=R} = \frac{K(T_r-T_\infty)R}{R^2} = \frac{K(T_r - T_\infty)}{R}$$

$$q_r \ni h\left(\cancel{T_r - T_\infty}\right) = \frac{K\left(\cancel{T_r - T_\infty}\right)}{R}$$

$$1 = \frac{hR}{K} = \frac{hD}{2K} \Rightarrow 2 = \frac{hD}{K}$$

d.) $Bi = \frac{h\cdot L}{K}$, $Nu = \frac{hD}{K}$

essentially the same thing

10B.2. Viscous heating in slit flow. Find the temperature profile for the viscous heating problem shown in Fig. 10.4-2, when given the following boundary conditions: at $x = 0$, $T = T_0$; at $x = b$, $q_x = 0$.

Answer: $\dfrac{T - T_0}{\mu v_b^2/k} = \left(\dfrac{x}{b}\right) - \dfrac{1}{2}\left(\dfrac{x}{b}\right)^2$

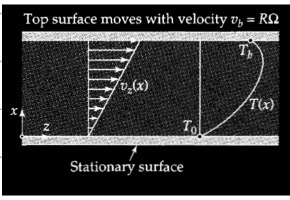


Fig. 10.4-2. Modification of a portion of the flow system in Fig. 10.4-1, in which the curvature of the bounding surfaces is neglected.

$q_x(x=b) = 0 \qquad T(x=0) = T_0 \qquad T(R-b) = T_0$

$T(R) = T_b$

$V_b = R\Omega$

$0 = -\int_S ds \langle q|n\rangle - \int_S ds \langle v|\pi|n\rangle \quad , \quad -\int_S ds \langle v|\pi|n\rangle = -\int_S ds\left(\langle v|\tau|n\rangle - \cancel{p\langle v|n\rangle}\right)$ 0 no pressure work $\Rightarrow \dfrac{[LW\Delta x\, v_z \tau_{zx}]_x - [LW\Delta x\, v_z \tau_{zx}]_{x+\Delta x}}{LW\Delta x}$

$V_z(x=b) = V_b = R\Omega$

$\tau_{zx} = -\mu\left(\dfrac{dv_z}{dx} + \cancel{\dfrac{dv_x}{dz}}\right)$ 0

$V_z = \dfrac{V_b}{b}x$

$\tau_{zx} = -\mu \dfrac{dv_z}{dx} = -\mu \dfrac{d}{dx}\left(\dfrac{V_b}{b}x\right) = -\mu\left(\dfrac{V_b}{b}\right)$

$-\int_S ds \langle q|n\rangle = \lim_{\Delta x \to 0} \dfrac{-\left([LWq_x]_x - [LWq_x]_{x+\Delta x}\right)}{LW\Delta x} = \dfrac{dq_x}{dx}$

$\dfrac{dq_x}{dx} = \mu\left(\dfrac{V_b}{b}\right)^2$

$\dfrac{d^2T}{dx^2} = -\dfrac{\mu}{k}\left(\dfrac{V_b}{b}\right)^2$

$\dfrac{dT}{dx} = -\dfrac{\mu}{k}\left(\dfrac{V_b}{b}\right)^2 x + C_1$

$T = -\dfrac{\mu}{2k}\left(\dfrac{V_b}{b}\right)^2 x^2 + C_1 x + C_2$

10B.4. Heat conduction in an annulus (Fig. 10B.4).

(a) Heat is flowing through an annular wall of inside radius r_0 and outside radius r_1. The thermal conductivity varies linearly with temperature from k_0 at T_0 to k_1 at T_1. Develop an expression for the heat flow through the wall.

(b) Show how the expression in (a) can be simplified when $(r_1 - r_0)/r_0$ is very small. Interpret the result physically.

Answer: **(a)** $Q = 2\pi L(T_0 - T_1)\left(\frac{k_0 + k_1}{2}\right)\left(\ln \frac{r_1}{r_0}\right)^{-1}$; **(b)** $Q = 2\pi r_0 L\left(\frac{k_0 + k_1}{2}\right)\left(\frac{T_0 - T_1}{r_1 - r_0}\right)$

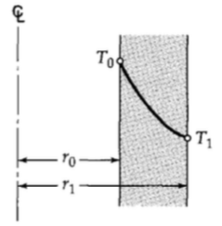


Fig. 10B.4. Temperature profile in an annular wall.

a.) $Q = Aq_r$

$m = \frac{K_1 - K_0}{T_1 - T_0}$ $T(r) = \frac{K_1 - K_0}{T_1 - T_0} r + T_0$

$Q = -\int_S dS \langle q|n\rangle = 2\pi(r_1 - r_0)^2 q_r$

$q_r = -K \frac{d}{dr}(T(r))$

$K = \frac{K_1 + K_0}{2}$

$q_r = -K \frac{d}{dr}(T)$

$\frac{Q}{2\pi r L} = -\left(\frac{K_1 + K_0}{2}\right)\frac{dT}{dr}$

$\int_{r_0}^{r_1} \frac{Q}{2\pi r L} dr = \int_{T_0}^{T_1} -\left(\frac{K_1 + K_0}{2}\right) dT$

$\frac{Q}{2\pi L} \ln\left(\frac{r_1}{r_0}\right) = (T_0 - T_1)\left(\frac{K_1 + K_0}{2}\right)$

$Q = 2\pi L (T_0 - T_1)\left(\frac{K_1 + K_0}{2}\right) \ln\left(\frac{r_1}{r_0}\right)^{-1}$

b.) when $\frac{(r_1 - r_0)}{r_0} << 0$

$\ln\left(\frac{r_1}{r_0}\right) = \frac{(r_1 - r_0)}{r_0}$

$Q = 2\pi L (T_0 - T_1)\left(\frac{K_1 + K_0}{2}\right) \frac{1}{\frac{(r_1 - r_0)}{r_0}}$

$Q = 2\pi r_0 L \frac{(T_0 - T_1)}{(r_1 - r_0)}\left(\frac{K_1 + K_0}{2}\right)$

✗

10B.7. Forced-convection heat transfer in flow between parallel plates (Fig. 10B.7). A viscous fluid with temperature-independent physical properties is in fully developed laminar flow between two flat surfaces placed a distance $2B$ apart. For $z < 0$ the fluid temperature is uniform at $T = T_1$. For $z > 0$ heat is added at a constant, uniform flux q_0 at both walls. Find the temperature distribution $T(x, z)$ for large z.

(a) Make a shell energy balance to obtain the differential equation for $T(x, z)$. Then discard the viscous dissipation term and the axial heat conduction term.

$q = \text{Constant}$

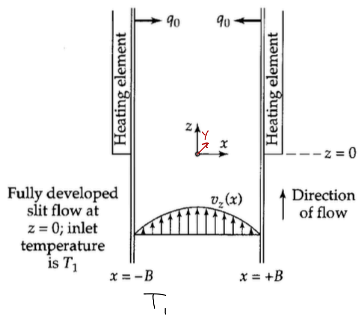


Fig. 10B.7. Laminar, incompressible flow between parallel plates, both of which are being heated by a uniform wall heat flux q_0 starting at $z = 0$.

(b) Recast the problem in terms of the dimensionless quantities

$$\Theta = \frac{T - T_1}{q_0 B/k} \qquad \sigma = \frac{x}{B} \qquad \zeta = \frac{kz}{\rho \hat{C}_p v_{z,\max} B^2} \tag{10B.7-1, 2, 3}$$

(c) Obtain the asymptotic solution for large z:

$$\Theta(\sigma, \zeta) = \frac{3}{2}\zeta + \frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 - \frac{39}{280} \tag{10B.7-4}$$

Assumptions $\quad T(x,z) = T_b(z) + \Psi(x)$

a.) · S.S.

- $V = V_z = V_z(x)$, $V_x = V_y = 0$
- $T = (x, z)$
- Constant - ρ, C_p, K, μ
  $q(x = \pm B) = q_0$
- $q_z \approx 0$
- $C_v = C_p \quad \frac{du}{dT} = C_p$
  $\Delta u = C_p \Delta T$
- No reversible work
- $|\tau| : |\nabla v| = -\mu\left(\frac{dv_z}{dx}\right)^2$

So we have Convection no build up LHS = 0

$$0 = -\int_S dS\, \rho\left(\tfrac{1}{2}v^2 + \phi + u\right)\langle v \cdot n\rangle - \int_S dS \langle q \cdot n\rangle - \int_S dS \langle v \cdot \tau \cdot n\rangle - \int_S dS\, \dot{s}$$
A $\quad$ B $\quad$ C

Only in the z $\quad$ no change in V_z

A: $-\int_S dS\, \rho\left(\cancel{\tfrac{1}{2}v^2} + \cancel{\phi} + u\right)$

$-\int_S dS\, u = -\int_S dS\, C_p \Delta T$

$$\frac{[W\Delta x \cdot \rho v_z C_p T(x,z)]_z - [W\Delta x\, \rho v_z C_p T(x,z)]_{z+\Delta z}}{W \Delta x \Delta z}$$
$$= \lim_{\Delta z \to 0} \rho v_z C_p \frac{T_z - T_{z+\Delta z}}{\Delta z} = \rho v_z C_p \frac{dT}{dz}$$

B: $-\int_S dS \langle q \cdot n\rangle = [2Bz\, q_0]_x - [2Bz\, q_0]_{x+\Delta x} + [2Bz\, q_z]_z - [2Bz\, q_z]_{z+\Delta z}$

$$= \frac{[z W q_0]_x - [z W q_0]_{x+\Delta x}}{\Delta x \Delta z\, W} + \frac{[x W z\, q_z]_z - [x W z\, q_z]_{z+\Delta z}}{\Delta z \Delta x\, W}$$
$$= \frac{\partial q_x}{\partial x} + \frac{\partial q_z}{\partial z}$$

C: $\int_S dS \langle v \cdot \tau \cdot n\rangle$

$-\int_S dS \left\{\langle v \cdot \tau \cdot n\rangle + \cancel{p \langle v \cdot n\rangle}\right\}$

$\tau_{zx} = -\mu \frac{dv_z}{dx}$

$$\frac{[W\Delta x \Delta z\, \tau_{zx} V_z]_x - [W \Delta z \Delta x\, \tau_{zx} V_z]_{x+\Delta x}}{W \Delta x \Delta z} = \left[W\Delta x \Delta z\, \mu\left(\frac{dV_z}{dx}\right)^2\right]$$

$q_x = -K\frac{\partial T}{\partial x}$, $q_z = -K\frac{\partial T}{\partial x}$

$$0 = \rho v_z C_p \frac{dT}{dz} + \frac{\partial q_x}{\partial x} + \frac{\partial q_z}{\partial z} + W\Delta x \Delta z\, \mu\left(\frac{dv_z}{dx}\right)^2$$

$$0 = \rho v_z C_p \frac{dT}{dz} + K\frac{d^2T}{dx^2} + \cancel{K\frac{d^2T}{dz^2}} + \cancel{W\Delta x \Delta z\, \mu\left(\frac{dv_z}{dx}\right)^2}$$

not significant part of Temp. Change most temp done on thursday $\quad$ Viscous dissipation term $\to 0$

a.) $\boxed{0 = \rho v_z C_p \frac{dT}{dz} - K\frac{d^2T}{dx^2}}$

$V_{z\max} = V_z(x = 0)$

energy gained from heating $\quad q_0 = -K\frac{dT}{dx}$

$\int_0^B q_0\, dx = \int_{T_1}^{T} -K\, dT \Rightarrow Bq_0 = -K(T - T_1)$ $\quad \underline{q_0 = \frac{-K(T-T_1)}{B}}$

@ $z = 0$, heat in $= \int_S dS \langle q \cdot n\rangle = 2\int_0^B dx \int_0^L dz\, \rho C_p (T - T_1) V_z$

$V_z(x = \pm B) = 0$

$V_z($

b.) $\rho v_z C_p \frac{dT}{dz} = K\frac{d^2T}{dx^2}\Big|_{x = \sigma B} \quad \frac{K}{B^2}\frac{d^2T}{d\sigma^2}$

$\rho v_z C_p \frac{d\left(\frac{T - T_1}{q_0 B/k}\right)}{dz} = \frac{K}{B^2}\frac{d^2\left(\frac{T-T_1}{q_0 B/k}\right)}{d\sigma^2}$, $\quad \Theta = \frac{T - T_1}{q_0 B/k} \qquad V_z = V_{z\max}\left[1 - \left(\frac{x}{B}\right)^2\right]$

$\rho v_z C_p \frac{d\Theta}{dz} = \frac{K}{B^2}\frac{d^2\Theta}{d\sigma^2}$ $\qquad V_z = V_{z\max}\left[1 - \left(\frac{x}{B}\right)^2\right]$

$\frac{d\Theta}{dz}\Big|_{z = \left(\frac{\rho C_p V_{z\max} B^2}{K}\right)\zeta} = \left(\frac{d\Theta}{d\zeta}\left(\frac{K}{\cancel{\rho C_p} V_{z\max} B^2}\right)(\cancel{\rho v_z C_p}) = \frac{K}{B^2}\frac{d^2\Theta}{d\sigma^2}\right)$

$\frac{d\Theta}{d\zeta}\left(\frac{\cancel{K}}{\cancel{B^2}}\left(\frac{\cancel{V_{z\max}}(1 - \left[\frac{x}{B}\right]^2)}{\cancel{V_{z\max}}}\right)\right) = \frac{\cancel{K}}{\cancel{B^2}}\frac{\partial^2\Theta}{\partial\sigma^2}\Big|_{\sigma = \frac{x}{B}}$

$\frac{d\Theta}{d\zeta}\left(1 - \left[\frac{x}{B}\right]^2\right) = \frac{\partial^2\Theta}{\partial\sigma^2}$ $\qquad \sigma = \frac{x}{B} \quad \zeta = \frac{Kz}{\rho C_p V_{z\max} B^2}$

$\boxed{\frac{d\Theta}{d\zeta}(1 - \sigma^2) = \frac{\partial^2\Theta}{\partial\sigma^2}}$

C.) $\Theta(\sigma, \zeta) = \frac{3}{2}\zeta + \frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 - \frac{39}{280}$

$T(r,z) = T_b(z) + \Psi(r) \qquad T(z,r)V_z(r) = T_b(z)V_z(r) + \Psi(r)V_z(r)$

$\Theta(\sigma, \zeta) = C\zeta + f(\sigma)$

$(1 - \sigma^2)\frac{\partial(C\zeta + f(\sigma))}{\partial\zeta} = \frac{\partial^2(C\zeta + f(\sigma))}{\partial\sigma^2}$

$C\int d\sigma\,(1 - \sigma^2) = \int \frac{\partial^2 f(\sigma)}{\partial\sigma^2}\, d\sigma$

$C_1 + C\left(\sigma - \frac{1}{3}\sigma^3\right) = \frac{\partial f(\sigma)}{\partial\sigma}$

$2\int_0^B \rho C_p V_z(x)(T_b - T_1)\, dx = 2q_0 z$ $\quad q_0 = -K\frac{dT}{dz}$ $\quad V_z(x) = V_{z\max}\left[1 - \left(\frac{x}{B}\right)^2\right]$

$T_b = T(x, z)$

$2\rho C_p V_{z\max}\int_0^B (T_b - T_1)\left(1 - \frac{x^2}{B^2}\right)dx = 2q_0 z$ $\quad x = \sigma B \quad dx = B\, d\sigma$ $\quad T - T_1 = \frac{\Theta q_0 B}{K}$ $\quad z = \zeta\left(\frac{\rho C_p V_{z\max} B^2}{K}\right)$

$\frac{q_0 B^2\, \cancel{2\rho C_p V_{z\max}}}{\cancel{K}}\int_0^1 \Theta(1 - \sigma^2)\, d\sigma = \left(\frac{2q_0\, \cancel{\rho C_p V_{z\max}} B^2}{\cancel{K}}\right)\zeta$

$\int_0^B \Theta(1 - \sigma^2)\, d\sigma = \zeta \qquad \frac{B}{B} = 1$

$\int_0^1 \Theta(1 - \sigma^2)\, d\sigma = \zeta$

$\zeta = \int_0^1 (1 - \sigma^2)[C\zeta + f(\sigma)]\, d\sigma$

$\frac{d}{d\zeta}\zeta = \left(C\zeta\int_0^1 (1 - \sigma^2)\, d\sigma + \int_0^1 f(\sigma)(1 - \sigma^2)\, d\sigma\right)\frac{d}{d\zeta}$

$1 = C\int_0^1 (1 - \sigma^2)\, d\sigma$

$1 = C\left(\sigma - \frac{1}{3}\sigma^3\Big|_0^1\right) \Rightarrow \quad 1 = \frac{2}{3}C$

$C = \frac{3}{2}$

$\Theta(\sigma, \zeta) = C\zeta + f(\sigma) = \frac{3}{2}\zeta + \frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 - \frac{39}{280}$

$\frac{\partial f(\sigma)}{\partial\sigma} = \frac{3}{2}\sigma - \frac{1}{2}\sigma^3 + C_1$

Assuming Symmetry: $\frac{\partial f(\sigma = 0)}{\partial\sigma} = 0$

So $C_1 = 0$

$f(\sigma) = \frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 - \frac{39}{280}$

$\Psi = f(\sigma) = \frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 + C_2$

$\int_0^1 f(\sigma)(1 - \sigma^2)\, d\sigma = \int_0^1 \left(\frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^3 + C_2\right)(1 - \sigma^2)\, d\sigma$

$0 = \int_0^1 \frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^3 + C_2 - \int_0^1 \left(\frac{3}{4}\sigma^4 - \frac{1}{8}\sigma^5 + C_2\sigma^2\right)d\sigma$

$0 = \frac{3}{12}(1) - \frac{1}{24} + C_2 - \left(\frac{3}{20}(1) - \frac{1}{48}(1) + \frac{1}{3}C_2\right)$

$\left(\frac{3}{20} - \frac{1}{48}\right) - \left(\frac{3}{12} - \frac{1}{24}\right) = \frac{2}{3}C_2$

$-\frac{39}{280} = C_2$

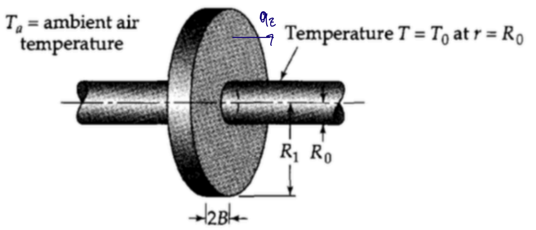


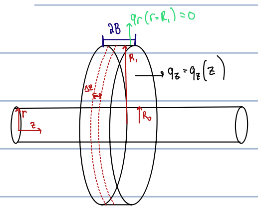
Fig. 10D.1. Circular fin on a heated pipe.

for rectangular $q_z \, S_A$ = height (=2B) x width (=W)

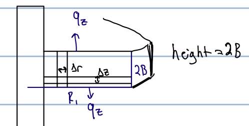
10D.1. Heat loss from a circular fin (Fig. 10D.1).

(a) Obtain the temperature profile $T(r)$ for a circular fin of thickness $2B$ on a pipe with outside wall temperature T_0. Make the same assumptions that were made in the study of the rectangular fin in §10.7.

(b) Derive an expression for the total heat loss from the fin.



So dimension



$q_z = h(T - T_a)$

a.) Assumptions: $T = T(r)$ $\quad T(r = R_0) = T_0$

T_a

$q_z = h(T_0 - T_a)$

$0 = -\int_S dS \langle q|n \rangle$

$$0 = \frac{\left[4\pi r B \, q_r\right]_r - \left[4\pi B r \, q_r\right]_{r+\Delta r} - 4\pi r B \Delta r \, h(T_0 - T_a)}{4\pi B \Delta r}$$

$$0 = -\int_S dS \langle q|n \rangle = \frac{\left[2\pi r (2B) q_r\right]_r - \left[2\pi r (2B) q_r\right]_{r+\Delta r} - 2\left(2\pi r \Delta r \, h (T - T_a)\right)}{2\pi \Delta r \, 2B}$$

$$0 = -\frac{1}{B} \lim_{\Delta r \to 0} \frac{[r q_r]_r - [r q_r]_{r+\Delta r}}{\Delta r} - \frac{h}{B} r (T - T_a)$$

$$0 = -\frac{1}{B}\frac{d}{dr}(r q_r) - \frac{h}{B} r (T - T_a)$$

$$0 = K \frac{d}{dr}\Big(\underbrace{r \frac{dT}{dr}}_{PR}\Big) - \frac{h}{B} r (T - T_a)$$

$$0 = \left(r \frac{d^2T}{\partial r^2} + \frac{dT}{dr}\right) - \frac{h}{KB} r (T - T_a) \qquad \Big| \; \Theta = T - T_a$$

$$0 = r \frac{d^2\Theta}{\partial r^2} + \frac{\partial \Theta}{\partial r} - \frac{h}{KB} r \Theta$$

$$0 = r^2 \frac{\partial^2\Theta}{\partial r^2} + r \frac{\partial \Theta}{\partial r} - \frac{h}{KB} r^2 \Theta$$

if we set $m^2 = \frac{h}{KB}$

$$0 = r^2 \frac{\partial^2\Theta}{\partial r^2} + r \frac{\partial \Theta}{\partial r} - (mr)^2 \Theta$$

$$\Theta = C_1 e^{N\xi} + C_2 e^{-N\xi}$$

$$\Theta = C_1 \sin(N\xi) + C_2 h \cos(N\xi)$$

b.) Q